

Construction of Gene Networks from Expression Profiles

Sriram Ponnambalam C
10405602

Guide : Prof. Srinivas Aluru

Outline

- ▶ Problem Definition, Gene Expression, and Gene Regulatory Networks.
- ▶ Methods
 1. Relevance Networks
 2. Gaussian Graphical Models
 3. Bayesian networks
 4. Module networks
- ▶ Performance

Outline

- ▶ Problem Definition, Gene Expression, and Gene Regulatory Networks.
- ▶ Methods
 1. Relevance Networks
 2. Gaussian Graphical Models
 3. Bayesian networks
 4. Module networks
- ▶ Performance

Problem Definition

- ▶ Biological Processes : Interactions between genes and gene products

Problem Definition

- ▶ Biological Processes : Interactions between genes and gene products
- ▶ Gene Regulatory network : Graph with Genes as vertices and edges representing interactions
- ▶ Input: Set of n genes $\{g_1, \dots, g_n\}$; m expressions measurements/gene
- ▶ Output: Gene Network G

Notations

- ▶ $\langle x_{i,1}, \dots, x_{i,m} \rangle$: Profile vector of g_i
- ▶ Expression profile matrix : Matrix D of size $n \times m$

Notations

- ▶ $\langle x_{i,1}, \dots, x_{i,m} \rangle$: Profile vector of g_i
- ▶ Expression profile matrix : Matrix D of size $n \times m$
- ▶ Random variable X_i : represent expression of a gene g_i
- ▶ $\mathcal{X} = \{X_1, \dots, X_n\}$

Gene Expression : Molecules

- ▶ DNA : Deoxyribonucleic acid
 - ▶ Carries genetic information
 - ▶ Double stranded sequence of nucleotides

Gene Expression : Molecules

- DNA : Deoxyribonucleic acid
 - Carries genetic information
 - Double stranded sequence of nucleotides
- Nucleotides
 - Bases: Adenine (A), Guanine(G), Cytosine(C),and Thymine(T)
 - A-T and G-C pairing

Gene Expression : Molecules

- ▶ DNA : Deoxyribonucleic acid
 - ▶ Carries genetic information
 - ▶ Double stranded sequence of nucleotides
- ▶ Nucleotides
 - ▶ Bases: Adenine (A), Guanine(G), Cytosine(C),and Thymine(T)
 - ▶ A-T and G-C pairing
- ▶ RNA : Single stand of nucleotides (Uracil instead of Thymine)

Gene Expression : Molecules

- DNA : Deoxyribonucleic acid
 - Carries genetic information
 - Double stranded sequence of nucleotides
- Nucleotides
 - Bases: Adenine (A), Guanine(G), Cytosine(C),and Thymine(T)
 - A-T and G-C pairing
- RNA : Single stand of nucleotides (Uracil instead of Thymine)
- Proteins
 - Polymers or Chains of Amino acids
 - 20 different types of Amino acids

Gene Expression ¹

DNA

5' - C - T - C - A - G - T - G - C - C - 3'
 3' - G - A - G - T - C - A - C - G - G - 5'

Protein

-Leucine — Serine — Alanine-

¹Pratt and Cornely (2004)

Gene Expression ¹

DNA

5' - C - T - C - A - G - T - G - C - C - 3'
3' - G - A - G - T - C - A - C - G - G - 5'

Transcription



mRNA

5' - C - U - C - A - G - U - G - C - C - 3'

Protien

—Leucine — Serine — Alanine—

¹Pratt and Cornely (2004)

Gene Expression ¹

DNA

5' - C - T - C - A - G - T - G - C - C - 3'
3' - G - A - G - T - C - A - C - G - G - 5'

Transcription



mRNA

5' - C - U - C - A - G - U - G - C - C - 3'

Translation



Protien

-Leucine — Serine — Alanine-

¹Pratt and Cornely (2004)

Micorarrays

- Devices that measure Expression Levels

Micorarrays

- Devices that measure Expression Levels
- mRNA levels $\propto$ Gene Expression; Measure mRNA levels

Micorarrays

- Devices that measure Expression Levels
- mRNA levels $\propto$ Gene Expression; Measure mRNA levels
- complementary-DNA arrays and oligonucleotide arrays

Gene Networks Properties - Zhu, Gerstein, and Snyder (2007)

- ▶ Gene Networks : Scale free distribution of degree

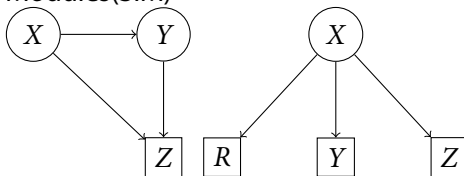
$$P(k) \propto k^{-\gamma}$$

Gene Networks Properties - Zhu, Gerstein, and Snyder (2007)

- Gene Networks : Scale free distribution of degree

$$P(k) \propto k^{-\gamma}$$

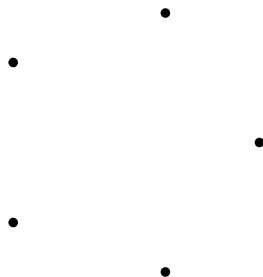
- Common motifs : Feed Forward loop(FFL), Single Input Modules(SIM)



Outline

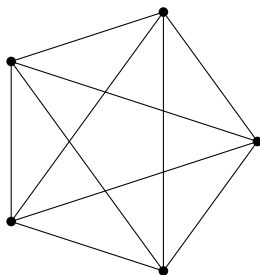
- Problem Definition, Gene Expression, and Gene Regulatory Networks.
- Methods
 1. Relevance Networks
 2. Gaussian Graphical Models
 3. Bayesian networks
 4. Module networks
- Performance

Relevance Networks



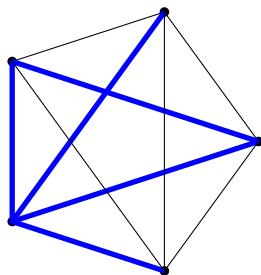
- ▶ Given n variable $X_1, \dots, X_n$ corresponding to $g_1, \dots, g_n$

Relevance Networks



- ▶ Given n variable $X_1, \dots, X_n$ corresponding to $g_1, \dots, g_n$
- ▶ Evaluate a relevance measure between each pair X_i, X_j

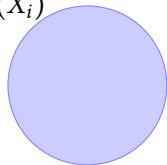
Relevance Networks



- ▶ Given n variable $X_1, \dots, X_n$ corresponding to $g_1, \dots, g_n$
- ▶ Evaluate a relevance measure between each pair X_i, X_j
- ▶ Add an edge if the measure is significant

Relevance Measure - Mutual Information

- Entropy of random variable

 $\mathcal{H}(X_i)$


$$\mathcal{H}(X) = \int P_X(x) \log P_X(x) dx$$

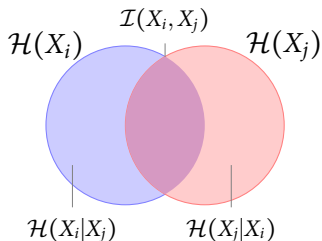
^aDaub et al., 2004

Relevance Measure - Mutual Information

- Entropy of random variable

$$\mathcal{H}(X) = \int P_X(x) \log P_X(x) dx$$

- Mutual Information $\mathcal{I}(X_i, X_j) = \mathcal{H}(X_i, X_j) - \mathcal{H}(X_i|X_j) - \mathcal{H}(X_j|X_i)$



Daub et al., 2004

MI Properties

- ▶ Invariant under homeomorphic transformations. Rank transformation is approx. homeomorphic.

MI Properties

- ▶ Invariant under homeomorphic transformations. Rank transformation is approx. homeomorphic.
 - ▶ $\langle x_{i,1}, \dots, x_{i,m} \rangle \rightarrow \langle \text{rank}(x_{i,1}), \dots, \text{rank}(x_{i,m}) \rangle$
 - ▶ Gene expression vector becomes a permutation of $\{1, 2, \dots, m\}$

MI Properties

- ▶ Invariant under homeomorphic transformations. Rank transformation is approx. homeomorphic.
 - ▶ $\langle x_{i,1}, \dots, x_{i,m} \rangle \rightarrow \langle \text{rank}(x_{i,1}), \dots, \text{rank}(x_{i,m}) \rangle$
 - ▶ Gene expression vector becomes a permutation of $\{1, 2, \dots, m\}$
- ▶ Data Processing Inequality(DPI) : For a Markov chain $X_i \rightarrow X_j \rightarrow X_k$

$$\mathcal{I}(X_i, X_k) \leq \mathcal{I}(X_i, X_j) \text{ and } \mathcal{I}(X_i, X_k) \leq \mathcal{I}(X_j, X_k)$$

MI Properties

- ▶ Invariant under homeomorphic transformations. Rank transformation is approx. homeomorphic.
 - ▶ $\langle x_{i,1}, \dots, x_{i,m} \rangle \rightarrow \langle \text{rank}(x_{i,1}), \dots, \text{rank}(x_{i,m}) \rangle$
 - ▶ Gene expression vector becomes a permutation of $\{1, 2, \dots, m\}$
- ▶ Data Processing Inequality(DPI) : For a Markov chain $X_i \rightarrow X_j \rightarrow X_k$

$$\mathcal{I}(X_i, X_k) \leq \mathcal{I}(X_i, X_j) \text{ and } \mathcal{I}(X_i, X_k) \leq \mathcal{I}(X_j, X_k)$$

- ▶ When a triple (X_i, X_j, X_k) satisfies DPI, the edge $X_i - X_k$ is removed.

Significance Testing - Hypothesis Testing²

1. Define a hypothesis H_0 : data is sampled from

$$P(X_i, X_j) = P(X_i)P(X_j)$$

²Koller and Friedman (2009)

Significance Testing - Hypothesis Testing²

1. Define a hypothesis H_0 : data is sampled from

$$P(X_i, X_j) = P(X_i)P(X_j)$$
2. Define a decision rule $R(D, t)$ for a relevance measure
 $MI(X_i, X_j, D)$ with a threshold t

²Koller and Friedman (2009)

Significance Testing - Hypothesis Testing²

1. Define a hypothesis H_0 : data is sampled from

$$P(X_i, X_j) = P(X_i)P(X_j)$$
2. Define a decision rule $R(D, t)$ for a relevance measure
 $MI(X_i, X_j, D)$ with a threshold t
3. Define $\mathcal{S}$: space of sufficient statistics $S[X_i, X_j]$ of m samples of
 X_i and X_j

²Koller and Friedman (2009)

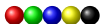
Significance Testing - Hypothesis Testing²

1. Define a hypothesis H_0 : data is sampled from
 $P(X_i, X_j) = P(X_i)P(X_j)$
2. Define a decision rule $R(D, t)$ for a relevance measure
 $MI(X_i, X_j, D)$ with a threshold t
3. Define $\mathcal{S}$: space of sufficient statistics $S[X_i, X_j]$ of m samples of
 X_i and X_j
 - p-value (t) : the false rejection probability when t is selected
for $R(D, t)$

$$P(S[X_i, X_j] | H_0, m)$$

²Koller and Friedman (2009)

Significance Testing - Permutation Testing³



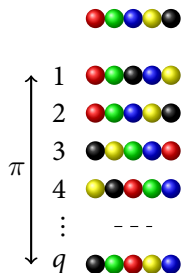
- ▶ Profile as a permutation vector

³Zola et al. (2010)

³Zola et al. (2010)

- ▶ Profile as a permutation vector
- ▶ Rule: Accept the dependency between X_i and X_j only if $\mathcal{I}(X_i, X_j) > \mathcal{I}(\pi(X_i), X_j)$ for at least $(1 - \epsilon)m!$ of $m!$ permutations

Significance Testing - Permutation Testing³

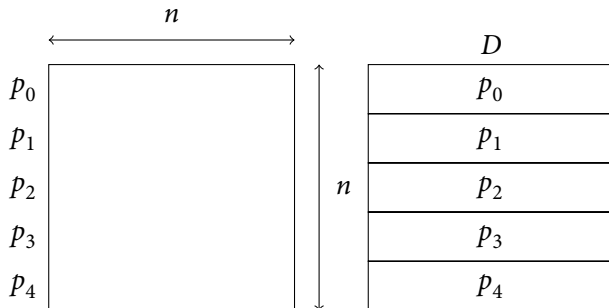


- ▶ Profile as a permutation vector
- ▶ Rule: Accept the dependency between X_i and X_j only if $\mathcal{I}(X_i, X_j) > \mathcal{I}(\pi(X_i), X_j)$ for at least $(1 - \epsilon)m!$ of $m!$ permutations
- ▶ Permute the profile vector $q < m!$ times

³Zola et al. (2010)

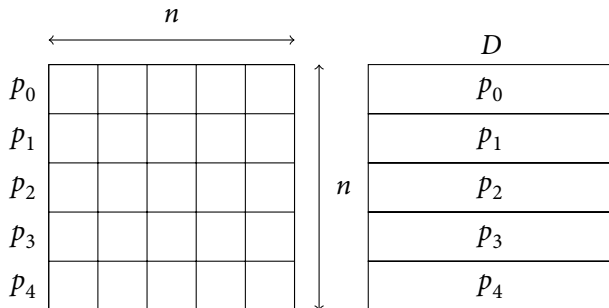
Parallel Algorithm - Zola et al. (2010) - Step 1

Compute MI for $q + 1$ pairs



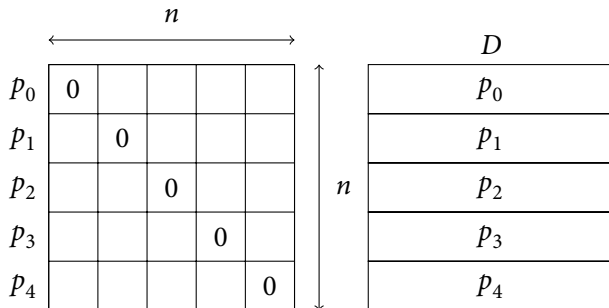
Parallel Algorithm - Zola et al. (2010) - Step 1

Compute MI for $q + 1$ pairs



Parallel Algorithm - Zola et al. (2010) - Step 1

Compute MI for $q + 1$ pairs



Parallel Algorithm - Zola et al. (2010) - Step 1

Compute MI for $q + 1$ pairs

Diagram illustrating the relationship between a 5x5 matrix and a 5x1 vector. The matrix has rows labeled p_0 to p_4 and columns labeled 0 to 4. A double-headed arrow above the matrix indicates its width is n . A vertical double-headed arrow to the right of the matrix indicates its height is n . To the right of the matrix is a vertical vector labeled D , with elements p_0 , p_1 , p_2 , p_3 , and p_4 .

Parallel Algorithm - Zola et al. (2010) - Step 1

Compute MI for $q + 1$ pairs

	0	1	2	3	4
p_0	0	1	2	3	4
p_1	1	0	2	3	4
p_2	2	2	0	3	4
p_3	3	3	3	0	4
p_4	4	4	4	4	0

Parallel Algorithm - Zola et al. (2010)

Bandwidth = $1/\mu$; Latency = ν ;

1. Compute MI for $q + 1$ pairs

- Compute Time : $O(qn^2m/p)$
- Communication Time : $O(\mu nm + p\nu)$

Parallel Algorithm - Zola et al. (2010)

Bandwidth = $1/\mu$; Latency = ν ;

1. Compute MI for $q + 1$ pairs

- Compute Time : $O(qn^2m/p)$
- Communication Time : $O(\mu nm + p\nu)$

2. Find MI value of rank $r = \epsilon \cdot q \cdot n(n - 1)/2$ by sorting, followed parallel reduction.

- Compute Time : $O((qn^2 \log n + r \log n)/p)$

Parallel Algorithm - Zola et al. (2010)

Bandwidth = $1/\mu$; Latency = ν ;

1. Compute MI for $q + 1$ pairs

- Compute Time : $O(qn^2m/p)$
- Communication Time : $O(\mu nm + p\nu)$

2. Find MI value of rank $r = \epsilon \cdot q \cdot n(n - 1)/2$ by sorting, followed parallel reduction.

- Compute Time : $O((qn^2 \log n + r \log n)/p)$

3. Use DPI is applied to eliminate indirect edges

- Compute Time : $O(n^3/p)$

Outline

- ▶ Problem Definition, Gene Expression, and Gene Regulatory Networks.
- ▶ Methods
 1. Relevance Networks
 2. Gaussian Graphical Models
 3. Bayesian networks
 4. Module networks
- ▶ Performance

Multivariate Gaussian

- ▶ D assumed to be drawn from a multivariate normal distribution $\mathcal{N}(\mathcal{X}, \mu, \Sigma)$ with $\mu = (\mu_1, \dots, \mu_n)^T$ and co-variance $\Sigma = (\sigma_{ij})$

Multivariate Gaussian

- ▶ D assumed to be drawn from a multivariate normal distribution $\mathcal{N}(\mathcal{X}, \mu, \Sigma)$ with $\mu = (\mu_1, \dots, \mu_n)^T$ and co-variance $\Sigma = (\sigma_{ij})$
- ▶ Partial correlation matrix $\Pi = (\pi_{ij})$: correlation between X_i and X_j conditioned on $\mathcal{X} - \{X_i, X_j\}$

Multivariate Gaussian

- ▶ D assumed to be drawn from a multivariate normal distribution $\mathcal{N}(\mathcal{X}, \mu, \Sigma)$ with $\mu = (\mu_1, \dots, \mu_n)^T$ and co-variance $\Sigma = (\sigma_{ij})$
- ▶ Partial correlation matrix $\Pi = (\pi_{ij})$: correlation between X_i and X_j conditioned on $\mathcal{X} - \{X_i, X_j\}$
- ▶ Notation $B = (\rho_{ij})$: Pearson correlation coefficient matrix;

$$\sigma_{ij} = \rho_{ij}\sigma_i\sigma_j$$

Standard GGM - $m \gg n$

1. Estimate unbiased sample covariance matrix

$$\hat{\Sigma} = (\hat{\sigma}_{ij}) = \frac{1}{m-1} \sum_{i=1}^m (D_{i,\cdot} - \hat{\mu})(D_{i,\cdot} - \hat{\mu})^T$$

$D_{i,\cdot}$ is the i -th row of D ; $\hat{\mu}$ is the sample mean vector

Standard GGM - $m \gg n$

1. Estimate unbiased sample covariance matrix

$$\hat{\Sigma} = (\hat{\sigma}_{ij}) = \frac{1}{m-1} \sum_{i=1}^m (D_{i,\cdot} - \hat{\mu})(D_{i,\cdot} - \hat{\mu})^T$$

$D_{i,\cdot}$ is the i -th row of D ; $\hat{\mu}$ is the sample mean vector

2. Compute partial correlation coefficients: With $\Omega = B^{-1} = (\omega_{ij})$,

$$\pi_{ij} = -\omega_{ij} / \sqrt{\omega_{ii}\omega_{jj}}$$

Can use Σ instead of B (since $\sigma_{ij} = \rho_{ij}\sigma_i\sigma_j$)

Standard GGM - $m \gg n$

1. Estimate unbiased sample covariance matrix

$$\hat{\Sigma} = (\hat{\sigma}_{ij}) = \frac{1}{m-1} \sum_{i=1}^m (D_{i,\cdot} - \hat{\mu})(D_{i,\cdot} - \hat{\mu})^T$$

$D_{i,\cdot}$ is the i -th row of D ; $\hat{\mu}$ is the sample mean vector

2. Compute partial correlation coefficients: With $\Omega = B^{-1} = (\omega_{ij})$,

$$\pi_{ij} = -\omega_{ij} / \sqrt{\omega_{ii}\omega_{jj}}$$

Can use Σ instead of B (since $\sigma_{ij} = \rho_{ij}\sigma_i\sigma_j$)

3. Use Hypothesis testing to compute significant correlations

Standard GGM - $m \gg n$

1. Estimate unbiased sample covariance matrix

$$\hat{\Sigma} = (\hat{\sigma}_{ij}) = \frac{1}{m-1} \sum_{i=1}^m (D_{i,\cdot} - \hat{\mu})(D_{i,\cdot} - \hat{\mu})^T$$

$D_{i,\cdot}$ is the i -th row of D ; $\hat{\mu}$ is the sample mean vector

2. Compute partial correlation coefficients: With $\Omega = B^{-1} = (\omega_{ij})$,

$$\pi_{ij} = -\omega_{ij} / \sqrt{\omega_{ii}\omega_{jj}}$$

Can use Σ instead of B (since $\sigma_{ij} = \rho_{ij}\sigma_i\sigma_j$)

3. Use Hypothesis testing to compute significant correlations
4. Visualize the inferred correlation structure by a graph.

Improvements - Schäfer and Strimmer (2005)

Improvements in GGM for $n > m$

- ▶ Moore-Penrose pseudo-inverse for Σ^{-1}

Improvements - Schäfer and Strimmer (2005)

Improvements in GGM for $n > m$

- ▶ Moore-Penrose pseudo-inverse for Σ^{-1}
- ▶ Bootstrap sampling improve upon unstable $\hat{\Sigma}^{-1}$

Improvements - Schäfer and Strimmer (2005)

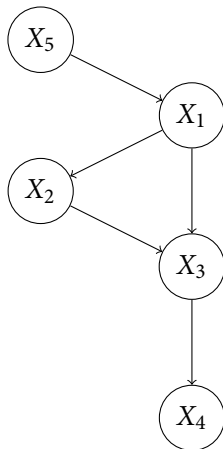
Improvements in GGM for $n > m$

- ▶ Moore-Penrose pseudo-inverse for Σ^{-1}
- ▶ Bootstrap sampling improve upon unstable $\hat{\Sigma}^{-1}$
- ▶ Hypothesis Testing using a mixture distribution

Outline

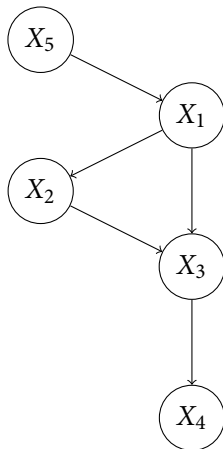
- ▶ Problem Definition, Gene Expression, and Gene Regulatory Networks.
- ▶ Methods
 1. Relevance Networks
 2. Gaussian Graphical Models
 3. Bayesian networks
 4. Module networks
- ▶ Performance

Bayesian Networks



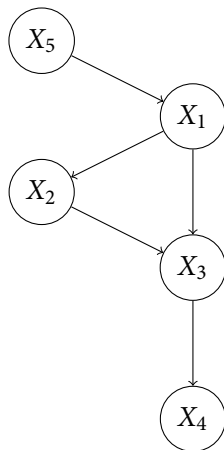
- ▶ BN (G, P) : G is a directed acyclic graph $(\mathcal{X}, E)$; P is the joint probability distribution of $\mathcal{X}$.

Bayesian Networks



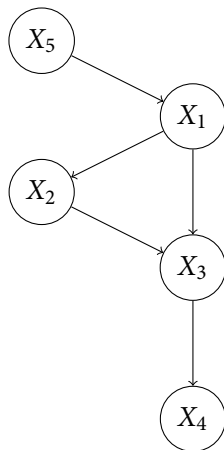
- ▶ $\text{BN}(G, P)$: G is a directed acyclic graph $(\mathcal{X}, E)$; P is the joint probability distribution of $\mathcal{X}$.
- ▶ *Parent* $\text{Pa}(X_i)$. Eg. X_1 is parent of X_2

Bayesian Networks



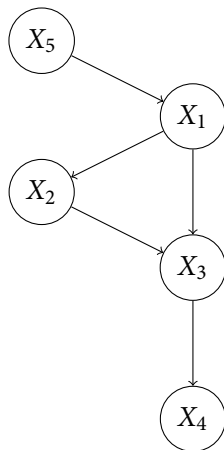
- ▶ $\text{BN}(G, P)$: G is a directed acyclic graph $(\mathcal{X}, E)$; P is the joint probability distribution of $\mathcal{X}$.
- ▶ *Parent* $\text{Pa}(X_i)$. Eg. X_1 is parent of X_2
- ▶ *Descendant* (X_i) . Eg. X_4 is a descendant of X_5

Bayesian Networks



- ▶ BN (G, P) : G is a directed acyclic graph $(\mathcal{X}, E)$; P is the joint probability distribution of $\mathcal{X}$.
- ▶ *Parent* $Pa(X_i)$. Eg. X_1 is parent of X_2
- ▶ *Descendant* (X_i) . Eg. X_4 is a descendant of X_5
- ▶ For X_i , given $Pa(X_i)$, X_i is independent of all its non-descendants. Eg. $X_3 \perp X_5$ given X_1, X_2

Bayesian Networks



- ▶ BN (G, P) : G is a directed acyclic graph $(\mathcal{X}, E)$; P is the joint probability distribution of $\mathcal{X}$.
- ▶ *Parent* $Pa(X_i)$. Eg. X_1 is parent of X_2
- ▶ *Descendant* (X_i) . Eg. X_4 is a descendant of X_5
- ▶ For X_i , given $Pa(X_i)$, X_i is independent of all its non-descendants. Eg. $X_3 \perp X_5$ given X_1, X_2
- ▶ Joint distribution

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i | Pa(X_i), \theta_i)$$

$$P(\mathcal{X}) = P(X_5)P(X_1|X_5)P(X_2|X_1)P(X_3|X_2, X_1)P(X_4|X_3)$$

Bayesian Networks Learning

Problem : Given D , construct G and estimate $P_{\mathcal{X}}$.

Bayesian Networks Learning

Problem : Given D , construct G and estimate $P_{\mathcal{X}}$.

- ▶ Need to learn both the structure and the parameters for each X_i ($\Theta_G = \{\theta_i, \dots, \theta_n\}$)
- ▶ General Case is NP-Hard (Chickering, 2004)

Bayesian Networks Learning

Problem : Given D , construct G and estimate $P_{\mathcal{X}}$.

- ▶ Need to learn both the structure and the parameters for each X_i ($\Theta_G = \{\theta_i, \dots, \theta_n\}$)
- ▶ General Case is NP-Hard (Chickering, 2004)
- ▶ Search through space of networks

Bayesian Networks Learning

Problem : Given D , construct G and estimate $P_{\mathcal{X}}$.

- ▶ Need to learn both the structure and the parameters for each X_i ($\Theta_G = \{\theta_i, \dots, \theta_n\}$)
- ▶ General Case is NP-Hard (Chickering, 2004)
- ▶ Search through space of networks
- ▶ Define a score proportional to $P(G|D)$ ($\text{score}(G, D) = \log P(G|D)$ is used normally)

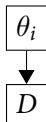
Bayesian Networks Learning

Problem : Given D , construct G and estimate $P_{\mathcal{X}}$.

- ▶ Need to learn both the structure and the parameters for each X_i ($\Theta_G = \{\theta_i, \dots, \theta_n\}$)
- ▶ General Case is NP-Hard (Chickering, 2004)
- ▶ Search through space of networks
- ▶ Define a score proportional to $P(G|D)$ (score(G, D) = $\log P(G|D)$ is used normally)
- ▶ Select a network with best score (maximum)

Graph Score - Bayesian Approach

- ▶ Consider $P(X_i|Pa(X_i); \theta_i)$; Estimate θ_i from D



Graph Score - Bayesian Approach

- ▶ Consider $P(X_i | Pa(X_i); \theta_i)$; Estimate θ_i from D
- ▶ Bayesian Approach: Assume a prior $P(\theta_i; \alpha_i)$

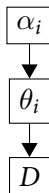
 α_i  θ_i  D

Graph Score - Bayesian Approach

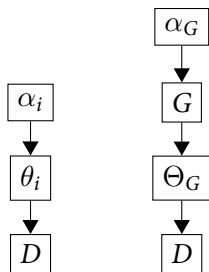
- ▶ Consider $P(X_i|Pa(X_i); \theta_i)$; Estimate θ_i from D
- ▶ Bayesian Approach: Assume a prior $P(\theta_i; \alpha_i)$
- ▶

$$P(\theta_i|D) \propto P(D|\theta_i)P(\theta_i)$$

$$\text{Estimate } \theta_i \text{ as } E_{P(\theta_i|D)}[\theta_i] = \int P(\theta_i|D)\theta_i d\theta_i$$



Graph Score - Bayesian Approach



- ▶ Consider $P(X_i|Pa(X_i); \theta_i)$; Estimate θ_i from D
- ▶ Bayesian Approach: Assume a prior $P(\theta_i; \alpha_i)$

$$P(\theta_i|D) \propto P(D|\theta_i)P(\theta_i)$$

Estimate θ_i as $E_{P(\theta_i|D)}[\theta_i] = \int P(\theta_i|D)\theta_i d\theta_i$

- ▶ Estimate $P(G|D)$ in the Bayesian approach

$$P(G|D) \propto P(D|G)P(G)$$

Graph Score - Decomposed

With certain independence assumptions (Heckerman, 1999),

$$\text{score}(G, D) = \log P(G|D) = \sum_i \text{LocalScore}(X_i, Pa(X_i), D)$$

Exact Learning - Ott, Imoto, and Miyano (2004)

- ▶ $O(n!2^{O(n^2)})$ possible DAGs

Exact Learning - Ott, Imoto, and Miyano (2004)

- ▶ $O(n!2^{O(n^2)})$ possible DAGs
- ▶ Define $F(X_i, A)$: Optimal score possible by choosing $Pa(X_i)$ from A

$$F(X_i, A) = \max \left\{ \begin{array}{l} \text{LocalScore}(X_i, A, D) \\ \max_{X_j \in A} F(X_i, A - \{X_j\}) \end{array} \right.$$

Exact Learning - Ott, Imoto, and Miyano (2004)

- ▶ $O(n!2^{O(n^2)})$ possible DAGs
- ▶ Define $F(X_i, A)$: Optimal score possible by choosing $Pa(X_i)$ from A

$$F(X_i, A) = \max \left\{ \begin{array}{l} \text{LocalScore}(X_i, A, D) \\ \max_{X_j \in A} F(X_i, A - \{X_j\}) \end{array} \right.$$

- ▶ Define $Q(A, \pi(A))$: Optimal score of network on A consistent with $\pi(A)$

$$Q(A, \pi(A)) = \sum_{X_i \in A} F(X_i, \{X_j | X_j \text{ precedes } X_i \text{ in } \pi(A)\})$$

Eg. $Q(\{X_1, X_2, X_3\}, \{X_2, X_3, X_1\}) =$
 $F(X_2, \emptyset) + F(X_3, \{X_2\}) + F(X_1, \{X_2, X_3\})$

Exact Learning - Ott, Imoto, and Miyano (2004)

- ▶ $O(n!2^{O(n^2)})$ possible DAGs
- ▶ Define $F(X_i, A)$: Optimal score possible by choosing $Pa(X_i)$ from A

$$F(X_i, A) = \max \begin{cases} \text{LocalScore}(X_i, A, D) \\ \max_{X_j \in A} F(X_i, A - \{X_j\}) \end{cases}$$

- ▶ Define $Q(A, \pi(A))$: Optimal score of network on A consistent with $\pi(A)$

$$Q(A, \pi(A)) = \sum_{X_i \in A} F(X_i, \{X_j | X_j \text{ precedes } X_i \text{ in } \pi(A)\})$$

Eg. $Q(\{X_1, X_2, X_3\}, \{X_2, X_3, X_1\}) =$
 $F(X_2, \emptyset) + F(X_3, \{X_2\}) + F(X_1, \{X_2, X_3\})$

- ▶ Define $M(A)$

$$M(A) = \arg \max_{\pi' \in \pi(A)} Q(A, \pi')$$

$M(\mathcal{X})$ is the optimal order.

Exact Learning - Algorithm - Ott, Imoto, and Miyano (2004)

1. Compute $F(X_i, A), \forall X_i, \forall A \subseteq \mathcal{X}$ in the order of A

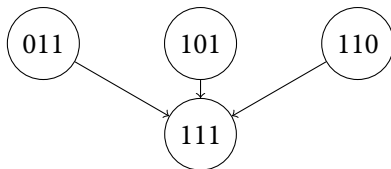
- $$\arg \min_{X_i \in A} \{F(X_i, A - \{X_i\}) + Q(A - \{X_i\}, M(A - \{X_i\}))\}$$



Exact Learning - Algorithm - Ott, Imoto, and Miyano (2004)

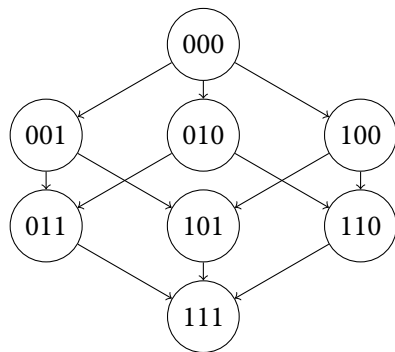
1. Compute $F(X_i, A), \forall X_i, \forall A \subseteq \mathcal{X}$ in the order of A
2. Compute $\forall X_i, \forall A \subseteq \mathcal{X}$ in the order of $|A| : M(A)[|A|]$

$$\arg \min_{X_i \in A} \{F(X_i, A - \{X_i\}) + Q(A - \{X_i\}, M(A - \{X_i\}))\}$$



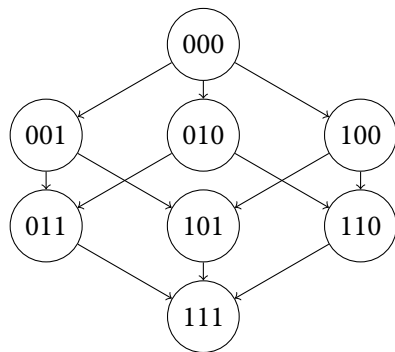
Complexity $O(n^2 \cdot 2^n)$

Exact Learning - Parallel Algorithm - Nikolova, Zola, and Aluru (2009)



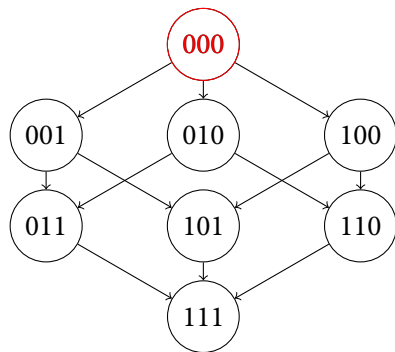
- Formulation on n -D hypercube (2^n processors): n steps

Exact Learning - Parallel Algorithm - Nikolova, Zola, and Aluru (2009)



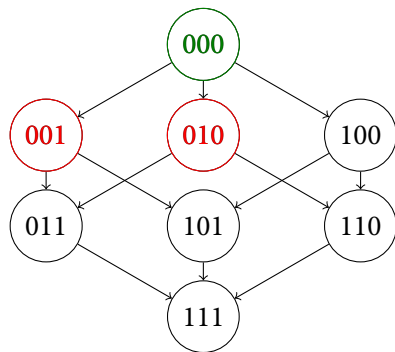
- ▶ Formulation on n -D hypercube (2^n processors): n steps
- ▶ 2^k processors available : 2^{n-k} times
- ▶ Pipelining hypercubes

Exact Learning - Parallel Algorithm - Nikolova, Zola, and Aluru (2009)



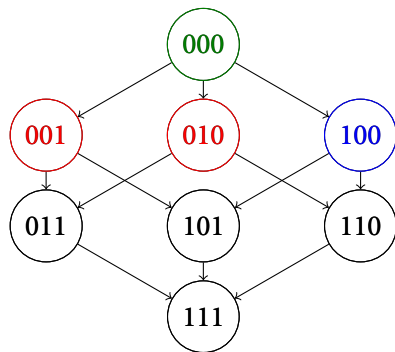
- ▶ Formulation on n -D hypercube (2^n processors): n steps
- ▶ 2^k processors available : 2^{n-k} times
- ▶ Pipelining hypercubes

Exact Learning - Parallel Algorithm - Nikolova, Zola, and Aluru (2009)



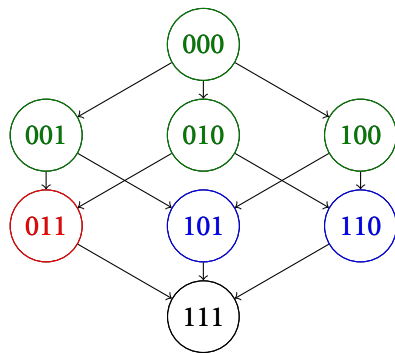
- ▶ Formulation on n -D hypercube (2^n processors): n steps
- ▶ 2^k processors available : 2^{n-k} times
- ▶ Pipelining hypercubes

Exact Learning - Parallel Algorithm - Nikolova, Zola, and Aluru (2009)

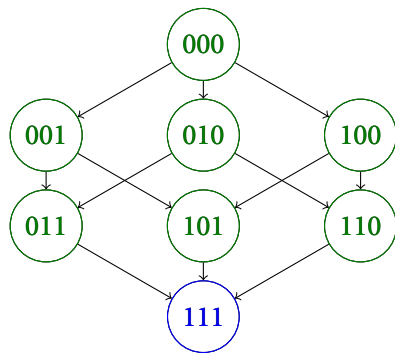


- ▶ Formulation on n -D hypercube (2^n processors): n steps
- ▶ 2^k processors available : 2^{n-k} times
- ▶ Pipelining hypercubes

Exact Learning - Parallel Algorithm - Nikolova, Zola, and Aluru (2009)



- ▶ Formulation on n -D hypercube (2^n processors): n steps
- ▶ 2^k processors available : 2^{n-k} times
- ▶ Pipelining hypercubes



- ▶ Formulation on n -D hypercube (2^n processors): n steps
- ▶ 2^k processors available : 2^{n-k} times
- ▶ Pipelining hypercubes
- ▶ Parallel Run-time : $O(n^2 \cdot 2^{n-k})$

Model Averaging

- ▶ Goal: Learn *features* that are common to the networks describing the data.

Model Averaging

- ▶ Goal: Learn *features* that are common to the networks describing the data.
- ▶ Features of Interest: Edge Feature, Parent Set Feature, Ancestor Relations

Model Averaging

- ▶ Goal: Learn *features* that are common to the networks describing the data.
- ▶ Features of Interest: Edge Feature, Parent Set Feature, Ancestor Relations
- ▶ Given a feature $f(G)$ of a graph,

$$E_{P(G|D)}[f(G)] = \sum_G f(G)P(G|D)$$

and

$$P(G|D) \propto \prod_i \text{LocalScore}(X_i, Pa(X_i), D)$$

Model Averaging

- ▶ Goal: Learn *features* that are common to the networks describing the data.
- ▶ Features of Interest: Edge Feature, Parent Set Feature, Ancestor Relations
- ▶ Given a feature $f(G)$ of a graph,

$$E_{P(G|D)}[f(G)] = \sum_G f(G)P(G|D)$$

and

$$P(G|D) \propto \prod_i \text{LocalScore}(X_i, \text{Pa}(X_i), D)$$

- ▶ Estimate feature by drawing T samples from $P(G|D)$

$$E_{P(G|D)}[f(G)] \approx \frac{1}{T} \sum_{t=1}^T f(G_t)$$

Model Averaging - Structure MCMC

- MCMC Sampling : Use a Markov chain with stationary distribution $P(G|D)$.

Model Averaging - Structure MCMC

- ▶ MCMC Sampling : Use a Markov chain with stationary distribution $P(G|D)$.
- ▶ $q(G|G')$: Markov chain's transition kernel

Model Averaging - Structure MCMC

- ▶ MCMC Sampling : Use a Markov chain with stationary distribution $P(G|D)$.
- ▶ $q(G|G')$: Markov chain's transition kernel
- ▶ Detailed Balance: If q satisfies (G, G' are two structures)

$$\frac{q(G|G')}{q(G'|G)} = \frac{P(G|D)}{P(G'|D)}$$

then the Markov chain $q(G|G')$ converges to $P(G|D)$.

Model Averaging - Structure MCMC ⁴

- ▶ Markov chain's transition kernel : Proposal distribution in Search Space

$$q(G|G') = \begin{cases} \frac{1}{|N(G')|} & \text{if } G \in N(G') \\ 0 & \text{otherwise} \end{cases}$$

$N(G')$ is the neighborhood of G' .

⁴Grzegorzcyk, Husmeier, and Werhli (2008)

Model Averaging - Structure MCMC ⁴

- ▶ Markov chain's transition kernel : Proposal distribution in Search Space

$$q(G|G') = \begin{cases} \frac{1}{|N(G')|} & \text{if } G \in N(G') \\ 0 & \text{otherwise} \end{cases}$$

$N(G')$ is the neighborhood of G' .

- ▶ Accept a neighbor G of G' with probability

$$A(G|G') = \min \left\{ 1, \frac{P(G|D)q(G'|D)}{P(G'|G)q(G|G')} \right\}$$

Equation of Detailed balance is satisfied.

⁴Grzegorzczyk, Husmeier, and Werhli (2008)

Model Averaging - Order MCMC - Friedman and Koller (2003)

Order MCMC adds two constraints to the search space

Model Averaging - Order MCMC - Friedman and Koller (2003)

Order MCMC adds two constraints to the search space

1. Structures are consistent with some known order π .

Model Averaging - Order MCMC - Friedman and Koller (2003)

Order MCMC adds two constraints to the search space

1. Structures are consistent with some known order π .
2. Number of parents any X_i can have is restricted to k

Model Averaging - Order MCMC - Friedman and Koller (2003)

Order MCMC adds two constraints to the search space

1. Structures are consistent with some known order π .
2. Number of parents any X_i can have is restricted to k

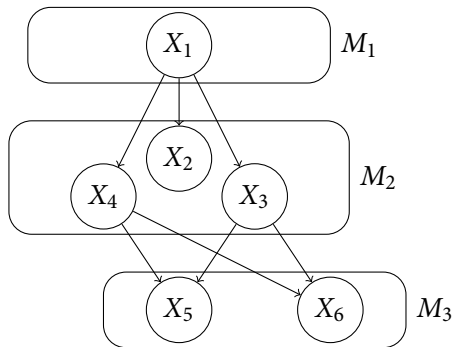
Samples are drawn from $P(f(G)|\pi, D)$ for a feature $f(G)$

$$P(f(G)|\pi, D) = \frac{P(f(G), D|\pi)}{P(D|\pi)}$$

Outline

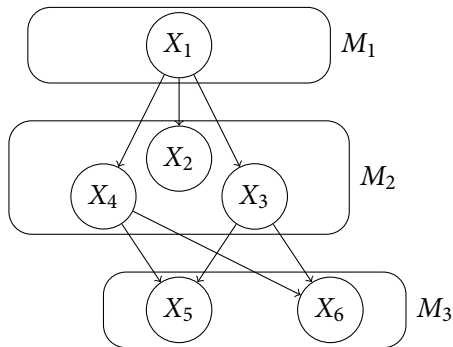
- ▶ Problem Definition, Gene Expression, and Gene Regulatory Networks.
- ▶ Methods
 1. Relevance Networks
 2. Gaussian Graphical Models
 3. Bayesian networks
 4. Module networks
- ▶ Performance

Module Network - Segal et al., 2003a



- ▶ Variables in the same module share the same set of parents and parameters

Module Network - Segal et al., 2003a



- ▶ Variables in the same module share the same set of parents and parameters
- ▶ $\mathcal{C} = \{M_1, \dots, M_k\}$: Partition of modules

Modules Network - Segal et al., 2003a

Goal: Learn $\mathcal{A}$, $\mathcal{S}$ and θ from the data D using a *score-based* approach

Modules Network Score - Segal et al., 2003a

- ▶ With certain assumptions

$$\text{score}(\mathcal{S}, \mathcal{A} : D) = \sum_{j=1}^K \text{score}_{M_j}(\text{Pa}(M_j), \mathcal{A}(X^j) : D)$$

Learning Algorithm - Segal et al., 2003a

1. *Structure Search Step*

- Learn $\mathcal{S}$ fixing $\mathcal{A}$
- Standard heuristic search over the space of structures
- Local search operators : Adding or removing X_i from a parent set $Pa(M_j)$.

2. *Module Assignment Search Step*

- Learns $\mathcal{A}$ fixing $\mathcal{S}$

$$\mathcal{A} = \arg \max_{\mathcal{A}'} \text{score}(\mathcal{S}, \mathcal{A}' : D)$$

- Compute the delta score for moving a variable from one module to another
- Iterate “Round-robin” fashion over all of variables, one at a time
 - until no single reassignment can improve the score

Outline

- ▶ Problem Definition, Gene Expression, and Gene Regulatory Networks.
- ▶ Methods
 1. Relevance Networks
 2. Gaussian Graphical Models
 3. Bayesian networks
 4. Module networks
- ▶ Performance

General Performance

- ▶ Zola et al. (2010) also extend the parallel algorithm for cell cluster, with which they constructed a 15,222 gene network for Arabidopsis
- ▶ Ma, Gong, and Bohnert (2007) used GGM to construct a network that covered the Arabidopsis genome
- ▶ Friedman and Koller (2003) found Order MCMC perform as good as Structure MCMC
- ▶ Segal et al. (2003b) constructed a module networks of 2,355 genes; Validated the hypotheses against gene annotations

Comparison - Grzegorczyk, Husmeier, and Werhli (2008)

- ▶ GGMs show a clearer separation of true and false edges than Relevance networks. Not necessarily improved performance.
- ▶ No significant difference between Bayesian networks and GGMs for observational data.
- ▶ For data from interventions, Bayesian networks outperform GGMs and Relevance networks.

Conclusion - Future Directions

- ▶ How to develop priors that can exploit scale-free nature of the gene regulatory networks ? Is it possible to use the power-law distribution as a prior for Bayesian networks ?
- ▶ How to use Bayesian networks for genome-wide gene network construction ? How to develop Bayesian network algorithms parallel ?
- ▶ How can we exploit modular nature of gene networks in developing parallel algorithms ?

Basso, Katia et al. (2005). “Reverse engineering of regulatory networks in human B cells”. In: *Nature Genetics* 37.4.

Chickering, David Maxwell (2004). “Large-Sample Learning of Bayesian Networks is NP-Hard”. In: *Journal of Machine Learning Research* 5, pp. 1287–1330.

Daub, Carsten O et al. (Aug. 2004). “Estimating mutual information using B-spline functions—an improved similarity measure for analysing gene expression data.” In: *BMC bioinformatics* 5, p. 118. ISSN: 1471-2105.

Friedman, Nir and Daphne Koller (2003). “Being Bayesian About Network Structure . A Bayesian Approach to Structure Discovery”. In: *Machine Learning*, pp. 95–125.

Grzegorczyk, Marco, Dirk Husmeier, and Adriano V. Werhli (2008). “Reverse Engineering Gene Regulatory Networks with Various Machine Learning Methods”. In: *Analysis of Microarray Data: A Network-Based Approach*. Ed. by Frank Emmert-Streib and Matthias Dehmer, pp. 100–141.

Heckerman, David (1999). "A tutorial on learning with Bayesian networks". In: Cambridge, MA, USA: MIT Press, pp. 301–354. ISBN: 0-262-60032-3.

Koller, Daphne and Nir Friedman (2009). *Probabilistic Graphical Models: Principles and Techniques*. The MIT Press.

Ma, Shisong, Qingqiu Gong, and Hans J Bohnert (Nov. 2007). "An Arabidopsis gene network based on the graphical Gaussian model." In: *Genome Research* 17.11, pp. 1614–25. ISSN: 1088-9051.

Nikolova, Olga, Jaroslaw Zola, and Srinivas Aluru (2009). "A Parallel Algorithm for Exact Bayesian Network Inference". In: *IEEE*, pp. 342–349.

Ott, S., S. Imoto, and S. Miyano (2004). "Finding Optimal Models for Small Gene Networks". In: *Pacific Symposium on Biocomputing*. Vol. 567, pp. 557–567.

Pratt, Charlotte W. and Kathleen Cornely (2004). *Essential Biochemistry*. John Wiley & Sons.

Schäfer, Juliane and Korbinian Strimmer (Mar. 2005). "An empirical Bayes approach to inferring large-scale gene association networks."

In: *Bioinformatics (Oxford, England)* 21.6, pp. 754–64. ISSN: 1367-4803.

Segal, Eran et al. (Dec. 2003a). “Learning Module Networks”. In: p. 10. ISSN: 0014-5793.

Segal, Eran et al. (June 2003b). “Module networks: identifying regulatory modules and their condition-specific regulators from gene expression data.” In: *Nature genetics* 34.2, pp. 166–76. ISSN: 1061-4036.

Zhu, Xiaowei, Mark Gerstein, and Michael Snyder (2007). “Getting connected : analysis and principles of biological networks”. In: *Genes & Development*, pp. 1010–1024.

Zola, Jaroslaw et al. (2010). “Parallel Information-Theory-Based Construction of Genome-Wide Gene Regulatory Networks”. In: *IEEE Transactions on Parallel and Distributed Systems* 21.12, pp. 1721–1733.